



Opioid Analgesia – Adults Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	Post Anaesthesia Unit (PACU) / Recovery Room	Medical and Nursing

2. Purpose

For post-operative patients within the Post Anaesthesia Care Unit (PACU) experiencing moderate to severe pain, in order to receive efficient and effective postoperative pain management and to reduce their pain to a comfortable level.

This is achieved by administering intravenous opioid in addition to other multimodal analgesia, administered in incremental doses titrated to the patient's pain score, level of sedation and vital signs.

This procedure is only for post-operative adult patients aged 18 years and over with body weight 50kg or above in the PACU / Recovery Room

3. Background

This procedure is developed to guide acute post-operative pain management only inside the PACU/recovery room of operation theatre complex/premises at AKG.

Other resources can be used in conjunction with opioids to assist in relieving pain post operatively. These may include explanations, reassurance, remaining with the patient to support them, re-positioning, elevation, and support of surgical site if appropriate, warming the patient if temperature is $< 36 (^{\circ} \text{C})$, non-steroidal anti-inflammatory drugs (preferably intravenous preparation for example parecoxib if no contraindications) and paracetamol.

4. Staff

- Registered Nurse (RN) – accredited to administer Intravenous Pain Protocol
- Medical officer – Consultant Anaesthetist or Anaesthesia/ICU Registrar

5. Equipment

- Opioid as per order on the Medication Chart (AKMR170B) in line with the relevant pain protocol
- Sodium Chloride 0.9%, syringe (10ml)
- Drawing up needle (blunt 18g needle)
- Appropriate labelling material – refer to the [National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines](#)
- Alcohol swabs

Before referencing this procedural document please ensure that you have the latest version from the [Policy Library](#) information hub page.

- Kidney dish

6. Clinical practice

6.1 Prescription

- Prescriber must adhere to AKG [Medication Management Procedure](#) - schedule 8 drugs
- The order is to be written on the back of the Medication Chart (AKMR170B) in the AS REQUIRED “PRN” MEDICATIONS section.
- The order must include Name of Drug, Route, Dose: “as per Recovery Room Pain Protocol.
- Standard AHS opiate concentrations
 - Morphine 10mg in 10mls
 - Fentanyl 100mcg in 10mls
 - Hydromorphone 2mg in 10mls
 - Morphine 10mg + Alfentanil 1mg in 10mls
 - Hydromorphone 2mg + Alfentanil 1mg in 10mls
- Unusual patient conditions and circumstances may lead to practice diverging from this procedure. This should be guided individually by the responsible Consultant Anaesthetist.

6.2 Patient assessment

The RN must assess the patient prior to commencing Pain Protocol for the following criteria:

- **Pain** – the location, cause, and severity of pain (appropriate to the surgery) using a numerical pain scale 0-10 or descriptors the patient can understand; mild, moderate or severe. Pain assessment may require the use of a multi-language visual analogue pain scale.
- **Sedation**- the patient must be fully awake or sedated but easily rousable
- **Respirations**- must be at a rate higher than 8 per minute
- **Blood pressure** – must be within 20% the patient’s normal limits using the preoperative blood pressure reading and patient history as a guidance

7. Procedure

- Check the IV cannula site for patency, swelling, inflammation
- 2 RNs to check the medication order and follow the AKG [Medication Management Procedure](#) for handling S8 drugs
- Draw up the prescribed opioid (Morphine 10mg, Fentanyl 100mcg, Hydromorphone 2mg, Morphine 1mg + Alfentanil 1mg, Hydromorphone 2mg + Alfentanil 1mg) into a 10ml syringe and make up to 10mls with Sodium Chloride 0.9%
- Label syringe with patient’s name, drug and concentration, date and time
- Refer to Opioid Pain Protocol Flow Chart (Appendix 1) to determine dose to administer
 - Adult 69 years or younger who is not frail or cachectic: 1 – 2ml every 5 minutes
 - Adult who is frail or cachectic or older than 69 years
0.5 - 1ml every 5 minutes
- 2 RNs to check the patient as per medication policy
- Ensure the patient is being administered oxygen via Hudson mask at 6 L/min

- Inject the first dose of opioid as a slow push and flush the line with the intravenous fluid in progress or 3-5ml of Sodium Chloride 0.9% drawn up in a separate syringe and labelled
- Record each dose on the medication chart as per medication policy
- Repeat the dose after a minimum of 5 minutes when sedation score is < 2 and Respiratory rate is no less than 8 per minute.
- Continue with incremental doses until the patient's pain score is 3/10 or less
- If the patient is not comfortable after giving 10ml of opioid (or maximum dose prescribed), obtain an anaesthetic review of the patient
- Patients with opioid resistant pain may require intravenous bolus Ketamine (100mg diluted in 10 ml normal saline = 10mg/ml) usually 10mg bolus at 5-10 minutes intervals. Maximum dose of 30 mg or 3mls.
- Please warn patients of possible agitation, visual hallucinations, or vivid dreams when ketamine is administered - better to dim the lights.
- Discontinue the pain protocol if sedation score is 2 or greater and obtain an anaesthetic review of the patient
- Discontinue the pain protocol if the respiration rate is 8 or less per minute and obtain an anaesthetic review of the patient

7.1 Observations

Once the pain protocol had been initiated the patient is to be monitored by the RN administering the opioid. At no time is the patient, or drug, to be left unattended.

- Record observations 10min: BP, Pulse, Respirations, Oxygen saturation, Sedation and Pain Scores
- Sedation level to be monitored continuously
- Pain Score to be assessed prior to each dose of the opioid
- All patients to be observed for a further 20 minutes (if up to 10ml opioid given) or 30 minutes (if greater than 10 ml opioid given) following the last dose of opioid before being assessed for readiness for discharge to the ward/unit
- Respiratory rate must be 10/minute or above, and sedation score of 0-1 at discharge

8. Hazards / Sub-optimal outcomes

- Over sedation. Sedation score of 2 – 3
- Respiratory depression. Respiratory rate less than 8 per minute.
- History of allergic reaction or an allergic reaction to the opioid in the recovery room. Patients who are unstable: hypotensive, hypovolemic, bradycardic, poor respiratory effort.
- **Assessment of the patient by the Anaesthetist is mandatory** in this situation prior to commencement of pain protocol.
- Concurrent epidural infusion, regional infusion and pain protocol/PCA:
- It is not routine for patients receiving an epidural infusion to also be administered intravenous pain protocol. Only in exceptional circumstances, following an anaesthetic review, will the anaesthetist order additional opioid via intravenous pain protocol.
- In these circumstances, clearly confirm with the Anaesthetist regarding the drug, doses and duration in order to avoid medication errors.

- Using the designated colour coded connection tubing and clear labelling is crucial – refer to the [National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines](#)
- Clear and concise handover using the ISOBAR minimum dataset should be completed with the ward staff upon discharge from the recovery

9. Educational notes

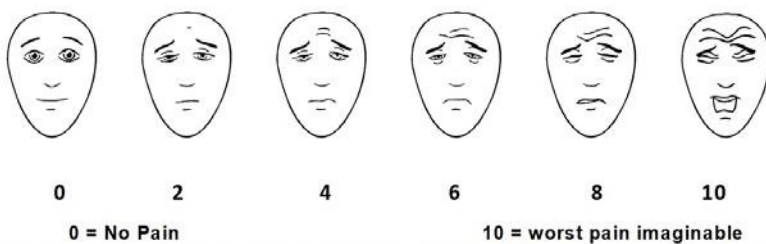
9.1 Numerical Rating Scale (NRS)

- In the NRS, the patient is asked to rate their pain from 0–10 with the understanding that 0 is equal to no pain and 10 is equal to the worst possible pain.
- A score of 1–3 can be considered mild pain, 4–6 as moderate pain and 7–10 as severe pain.

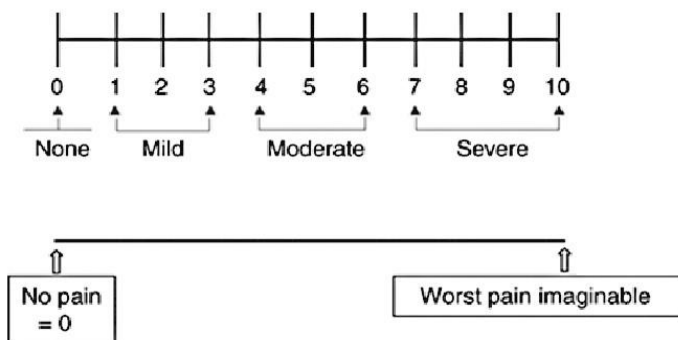
9.2 Visual Analogue Scale (VAS) / Faces Pain Scale (FPS)

If the patient is unable to verbalize, VAS or FPS can be used to assess pain intensity – see the scales below:

Faces Pain Scale (FPS)



Visual Analogue Scale (VAS)



In VAS and FPS scale, scoring of 4–6 is considered moderate pain and beyond the score of 6 is considered severe pain.

- Faces, Legs, Activity, Cry and Consolability (FLACC) scale

Adult patients who have difficulty communicating their pain (e.g., patients with cognitive impairment or who are critically unwell) require special attention as do patients whose language or cultural background differs significantly from that of their health care team. Communication

aids and behavioural scales such as the modified Faces, Legs, Activity, Cry and Consolability (FLACC) scale can be used in these situations.

If required seek assistance from an Interpreter Service – refer to the [Interpreter and Translator Guidelines](#).

Categories	0	1	2	Total
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disorientated	Frequent to constant frown, quivering chin, clenched jaw	
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking or legs drawn up	
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid or jerking	
Cry	No Cry (awake or asleep)	Moans or whimpers, occasional complaints	Crying steadily, scream or sobs, frequent complaints	
Consolability	Content, relaxed	Reassured by occasional touching, or being talked to; distractible	Difficult to console or comfort	

9.3 Functional Activity Score

- Is pain limiting function?
- Steps
 - Pain at rest (0 to 10)
 - Pain during activity (0 to 10), e.g., deep breathing, getting out bed
- Score
 - A: No limitation
 - B: Mild-moderate limitation
 - C: Severe limitation

Sedation Score

0 = Awake, Alert

1 = Easy to rouse, remains awake

2 = Easy to rouse unable to remain awake

3 = Difficult to rouse

1. A score of 2 is taken to indicate early opioid-induced ventilatory impairment (OIVI) and therefore the aim should be to titrate an opioid so that a patient's sedation score is always less than 2.
2. Note that a sedation score (e.g., 'sedation score less than 2') may be specified in the 'Max Dose/24 hrs' in the PRN section of the ACSQHC *National Inpatient Medication Chart* (NIMC) to indicate the maximum amount to be administered in 24 hrs when prescribing opioids in (Australian Commission on Safety and Quality in Health Care, 2019).

- RN accreditation to administer pain protocol consists of successful completion of a worksheet and a competency assessment
- Fentanyl, morphine or morphine/alfentanil are preferred opioids used at Armadale Health Service, hydromorphone and hydromorphone/alfentanil are occasionally prescribed.
- Reduced dosing is generally required in patients with renal or liver disease and patients over 70 years of age owing to altered metabolism and excretion of opioids.

10. Compliance monitoring

Compliance against this procedure will be monitored by the Opioid Stewardship Sub-Committee.

11. Policy context

The following documents are required to give effect to this procedure:

- *Medicines and Poisons Act*
- HDWA [High Risk Medication Policy MP 0131/20](#)
- HDWA [Guidelines for Managing Specific High Risk Medication Relevant to the Organisation](#)
- AKG [Medication Policy](#)
- AKG [Medication Management Procedure](#)
- AKG [Interpreter and Translator Guideline](#)

12. Supporting information

The following documents support the implementation of this procedure:

- Medication Chart AKMR 170B
- Perioperative Nursing Care Plan and Record AKMR93

13. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Medication Safety
- Standard 6 – Communicating for Safety Standard

References

1. Hatfield, A., & Tronson, M. (2009). The complete recovery room book (4th ed.). Oxford: Oxford University Press.
2. Schug, S. A., Palmer, G. M., Scott, A. D., Halliwell, R., & Trinca, J. Acute pain management: scientific evidence, fourth edition (2016) <http://www.anzca.edu.au/resources/college-publications>
3. Myles, P. S., Myles, D. B., Gallagher, W., Boyd, D., Chew, C., MacDonald, N., & Dennis, A. (2017). Measuring acute postoperative pain using the visual analog scale: the minimal clinically important difference and patient acceptable symptom state. British Journal of Anaesthesia, 118(3), 424-429. <http://dx.doi.org/10.1093/bja/aew466>
4. International Association for the study of pain (IASP) education material and resources: <http://www.iasp-pain.org/Resources?navItemNumber=650>
5. Therapeutic Guidelines Australia (2020): https://tgldcdp-tg-org-au.smhslibresources.health.wa.gov.au/viewTopic?etgAccess=true&guidelinePage=Pain%20and%20Analgesia&topicfile=severe-acute-nociceptive-pain&guidelinename=Pain%20and%20Analgesia§ionId=toc_d1e447#toc_d1e447

14. Document contact

Enquiries relating to this procedure may be directed to:

Title: Consultant Anaesthetist
 Department: Anaesthesia
 Email: raj.potula@health.wa.gov.au

15. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	05/07/2023	05/07/2026	Scheduled review

16. Authorisation

This procedure has been authorised and issued by:

Authorised by	Director of Clinical Services
Authorisation date	04/07/2023
Published date	05/07/2023

Appendix 1:

Preparing Opioids for PACU/Recovery Room Pain Protocol for Adults

The following instructions for preparation of opioids are to be followed for the adult Recovery Room Pain Protocol Flowchart for Adults (Appendix 2) in liaison with the above protocol.

Fentanyl protocol

100micrograms of fentanyl in 2ml diluted with 8ml of sodium chloride 0.9% to make 10ml of 10micrograms/ml of fentanyl.

Morphine protocol

10mg of morphine sulphate in 1ml diluted with 9ml of sodium chloride 0.9% to make 10ml of 1mg/ml of morphine.

Hydromorphone protocol

2mg of hydromorphone in 1.0 ml diluted with 9 ml of sodium chloride 0.9% to make 200micrograms/ml of hydromorphone.

Morphine and Alfentanil protocol

10mg of morphine sulphate in 1ml and 1mg (1000 microgram) of alfentanil in 2ml diluted with 7ml of sodium chloride 0.9% to make 10ml of 1mg /100 mcg of Morphine/Alfentanil mix per 1 ml.

Hydromorphone and Alfentanil protocol

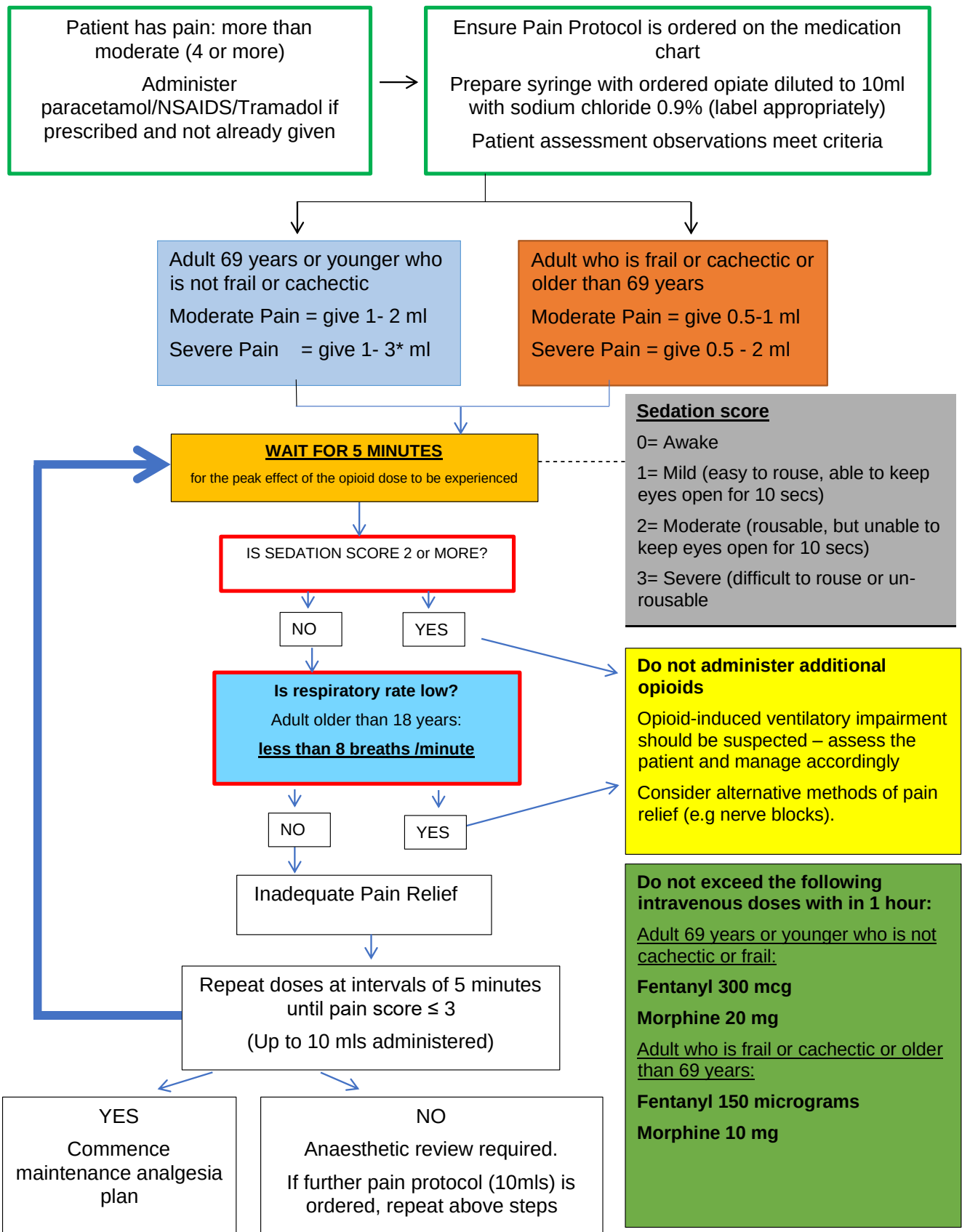
2mg of hydromorphone in 1.0 ml and 1mg (1000microgram) of alfentanil in 2ml is diluted with 7 ml of sodium chloride 0.9% to make 10 ml of 1mg/100mcg of hydromorphone/alfentanil mix per 1 ml.

Note:

- This preparation guide is based on the current pharmacy supply and may need to be re-calculated and modified accordingly if the dose and concentration of supply changes.
- The remaining drug in the syringe need to be handled according to [Medication Management Procedure](#) and kept sterile with sterile caps till the protocol is completed and medication are discarded as per [Medication Management Procedure](#) requirements.

Appendix 2:

PACU/Recovery Room Pain Protocol Flow Chart for Adults



*Except for Morphine/Alfentanil, Hydromorphone & Hydromorphone/Alfentanil where max dose is 2ml